



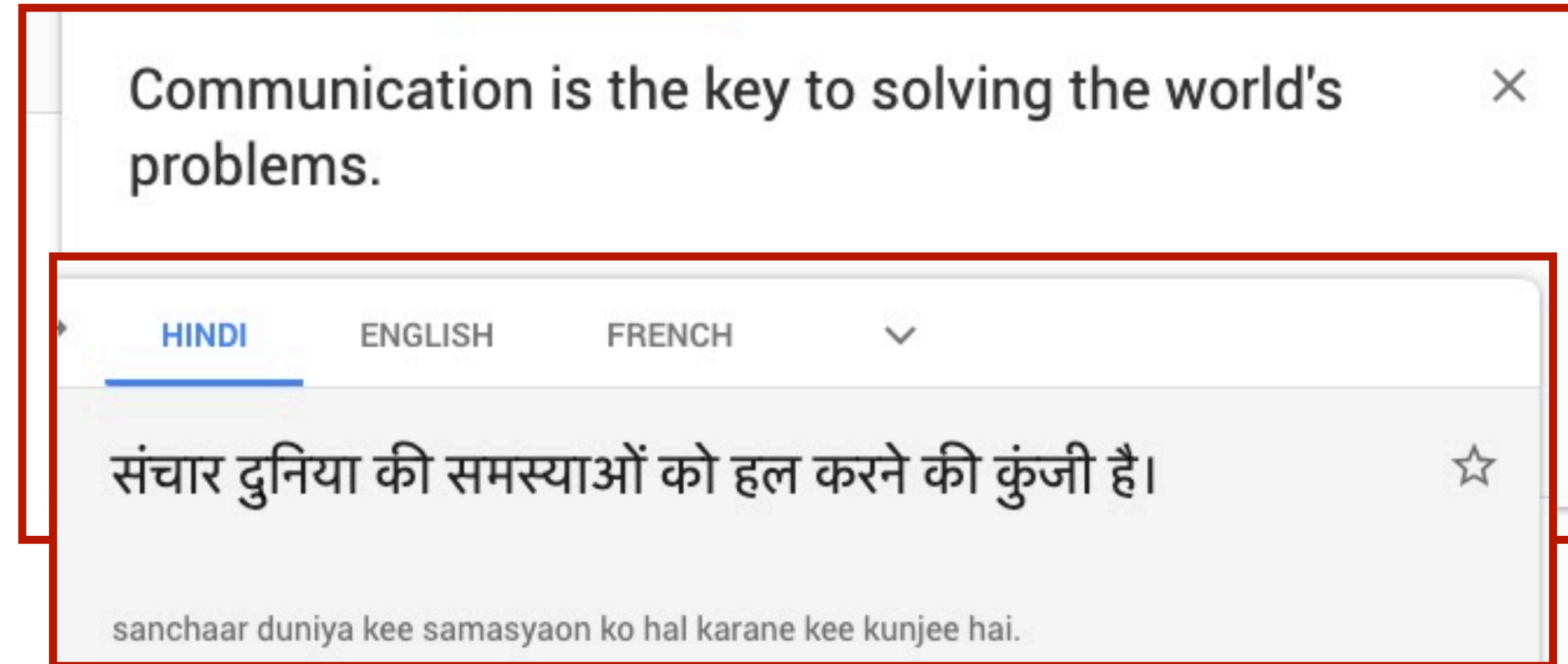
CMPT 825: Natural Language Processing

Machine Translation

Spring 2020

Adapted from slides from Chris Manning, Abigail See, Matthew Lamm,
Danqi Chen and Karthik Narasimhan

Translation



- One of the “holy grail” problems in artificial intelligence
- Practical use case: Facilitate communication between people in the world
- Extremely challenging (especially for low-resource languages)

Easy and not so easy translations

- Easy:
 - I like apples ↔ ich mag Äpfel (German)
- Not so easy:
 - I like apples ↔ J'aime les pommes (French)
 - I like red apples ↔ J'aime les pommes rouges (French)
 - *les* ↔ *the* but *les pommes* ↔ *apples*

MT basics

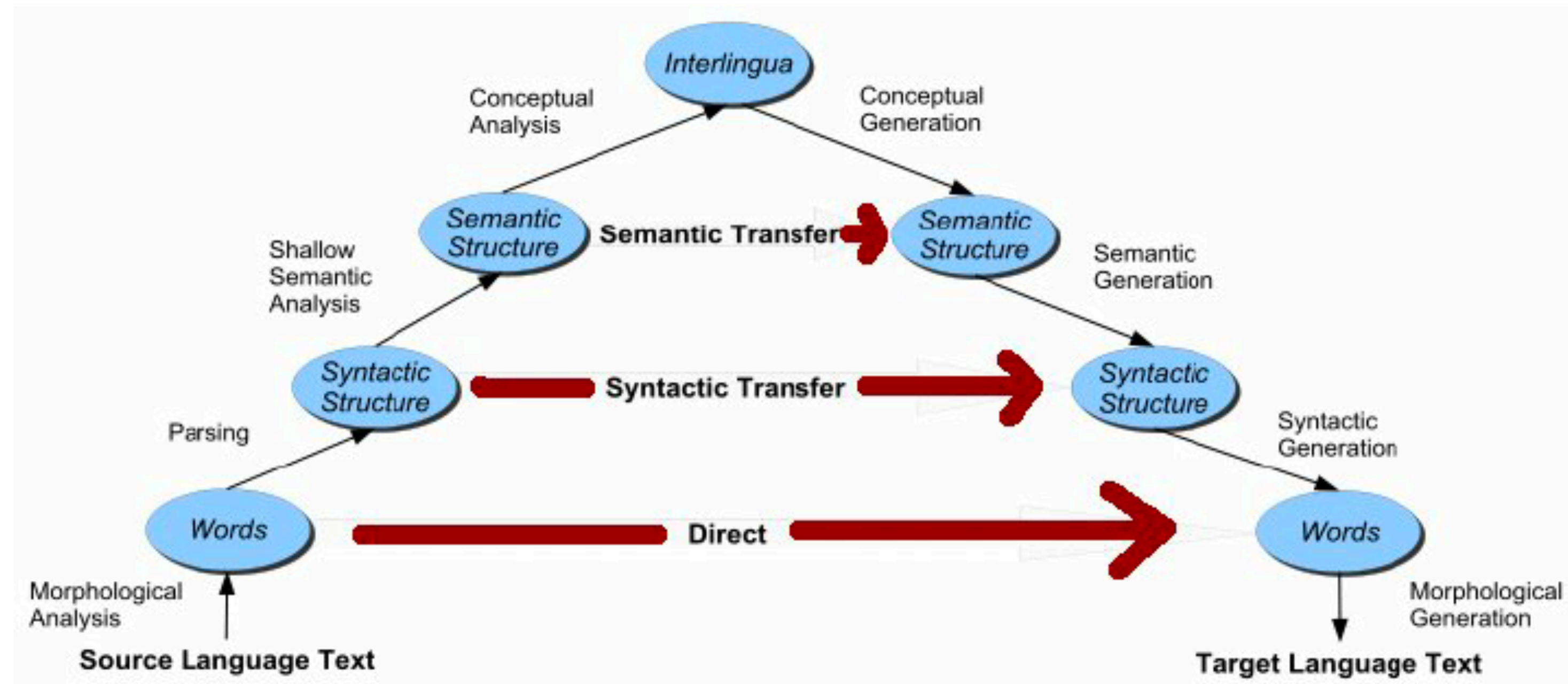
- **Goal:** Translate a sentence $w^{(s)}$ in a **source language (input)** to a sentence in the **target language (output)**
- Can be formulated as an optimization problem:
 - $\hat{w}^{(t)} = \arg \max_{w^{(t)}} \psi (w^{(s)}, w^{(t)})$
 - where ψ is a scoring function over source and target sentences
- Requires **two** components:
 - Learning algorithm to compute parameters of ψ
 - Decoding algorithm for computing the best translation $\hat{w}^{(t)}$

Why is MT challenging?

- Single words may be replaced with multi-word phrases
 - I like **apples** $\leftrightarrow$ J'aime **les pommes**
- Reordering of phrases
 - I like **red apples** $\leftrightarrow$ J'aime **les pommes rouges**
- Contextual dependence
 - *les* $\leftrightarrow$ *the* but *les pommes* $\leftrightarrow$ *apples*

Extremely large output space $\implies$ Decoding is NP-hard

Vauquois Pyramid



- Hierarchy of concepts and distances between them in different languages
- Lowest level: individual words/characters
- Higher levels: syntax, semantics
- Interlingua: Generic language-agnostic representation of meaning

Evaluating translation quality

- Two main criteria:
- **Adequacy:** Translation $w^{(t)}$ should adequately reflect the linguistic content of $w^{(s)}$
- **Fluency:** Translation $w^{(t)}$ should be fluent text in the target language

	Adequate?	Fluent?
<i>To Vinay it like Python</i>	yes	no
<i>Vinay debugs memory leaks</i>	no	yes
<i>Vinay likes Python</i>	yes	yes

Different translations of *A Vinay le gusta Python*

Evaluation metrics

- Manual evaluation is most accurate, but expensive
- Automated evaluation metrics:
 - Compare system hypothesis with reference translations
 - BiLingual Evaluation Understudy (BLEU) (Papineni et al., 2002):
 - Modified n -gram precision

$$p_n = \frac{\text{number of } n\text{-grams appearing in both reference and hypothesis translations}}{\text{number of } n\text{-grams appearing in the hypothesis translation}}$$

BLEU

$$\text{BLEU} = \exp \frac{1}{N} \sum_{n=1}^N \log p_n$$

Two modifications:

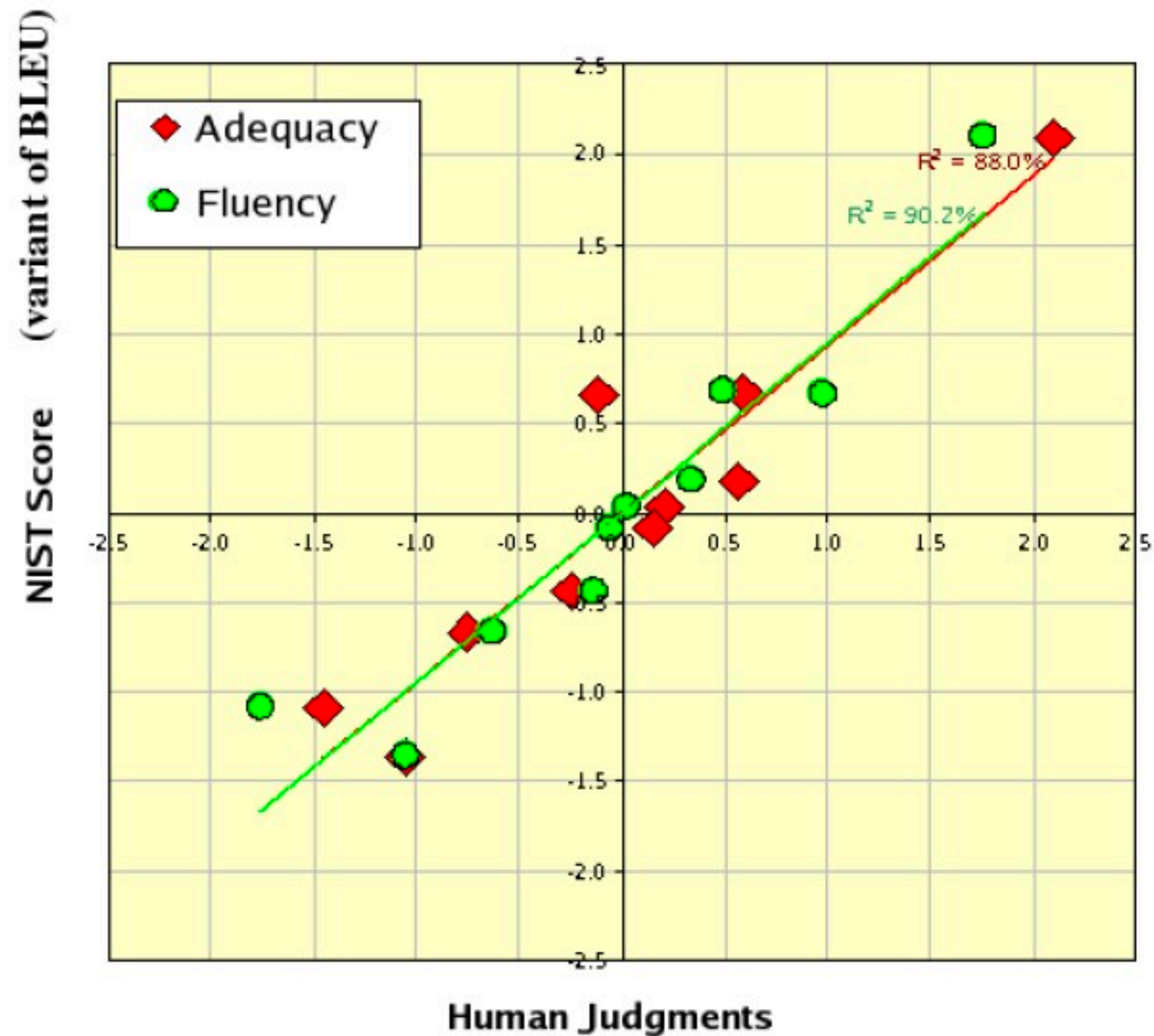
- To avoid $\log 0$, all precisions are smoothed
- Each n-gram in reference can be used at most once
 - Ex. **Hypothesis:** *to to to to to* vs **Reference:** *to be or not to be*
should not get a unigram precision of 1

Precision-based metrics favor short translations

- Solution: Multiply score with a brevity penalty for translations shorter than reference, $e^{1-r/h}$

BLEU

- Correlates somewhat well with human judgements



(G. Doddington, NIST)

BLEU scores

	Translation	p_1	p_2	p_3	p_4	BP	BLEU
<i>Reference</i>	<i>Vinay likes programming in Python</i>						
<i>Sys1</i>	<i>To Vinay it like to program Python</i>	$\frac{2}{7}$	0	0	0	1	.21
<i>Sys2</i>	<i>Vinay likes Python</i>	$\frac{3}{3}$	$\frac{1}{2}$	0	0	.51	.33
<i>Sys3</i>	<i>Vinay likes programming in his pajamas</i>	$\frac{4}{6}$	$\frac{3}{5}$	$\frac{2}{4}$	$\frac{1}{3}$	1	.76

Sample BLEU scores for various system outputs

Issues?

- Alternatives have been proposed:
- METEOR: weighted F-measure
- Translation Error Rate (TER): Edit distance between hypothesis and reference

Machine Translation (MT)

task of translating a sentence
from one language (the **source language**)
to a sentence in another language (the **target language**)

History

- Started in the **1950s**: rule-based, tightly linked to formal linguistics theories
 - Russian → English (motivated by the Cold War!)
 - Systems were mostly **rule-based**, using a bilingual dictionary to map Russian words to their English counterparts



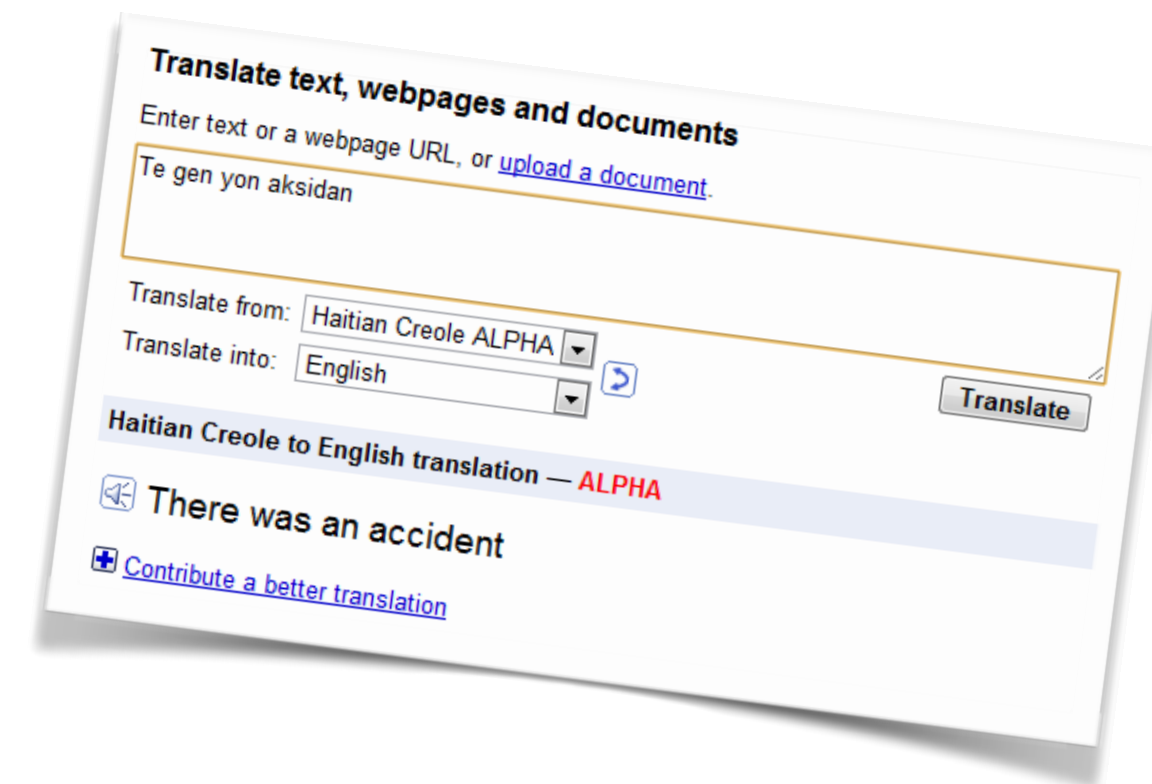
1 minute video showing 1954 MT:
<https://youtu.be/K-HfpsHPmvw>

History

- Started in the 1950s: rule-based, tightly linked to formal linguistics theories
- (late) 1980s to 2000s: Statistical MT
- 2000s-2014: Statistical Phrase-Based MT
- 2014-Present: Neural Machine Translation

History

- Started in the 1950s: rule-based, tightly linked to formal linguistics theories
- 1980s: Statistical MT
- 2000s-2015: **Statistical Phrase-Based MT**
- 2015-Present: Neural Machine Translation



Statistical MT

- Key Idea: Learn **probabilistic model** from **data**
- To find the best English sentence **y**, given French sentence **x**:

$$\hat{y} = \arg \max_y P(y | x)$$

- Decompose using Bayes Rule:

$$\hat{y} = \arg \max_y P(x | y) P(y)$$

Translation/Alignment Model

Models how words and phrases should be translated (*adequacy/fidelity*).

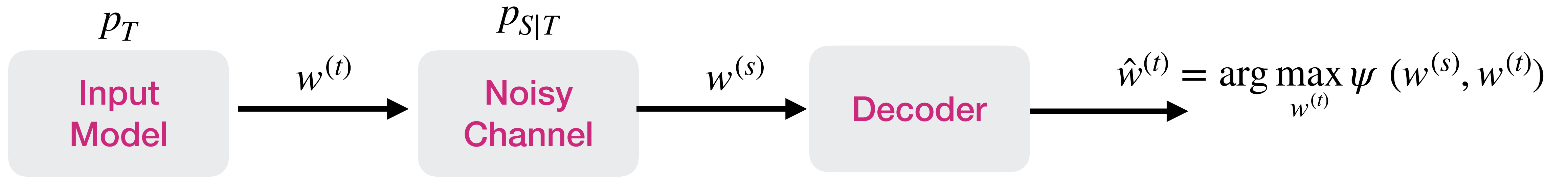
Learn from *parallel* data

Language Model

Models how to write good English (*fluency*)

Learn from *monolingual* data

Noisy channel model



$$\psi(w^{(s)}, w^{(t)}) = \psi_A(w^{(s)}, w^{(t)}) + \psi_F(w^{(t)})$$

$$\log P_{S,T}(w^{(s)}, w^{(t)}) = \log P_{S|T}(w^{(s)} | w^{(t)}) + \log P_T(w^{(t)})$$

- Generative process for source sentence
- Use Bayes rule to recover $w^{(t)}$ that is maximally likely under the conditional distribution $p_{T|S}$ (which is what we want)

Data

- Statistical MT relies requires lots of **parallel corpora**

1. Chapter 4, Koch (DE)	de	es
context We would like to ensure that there is a reference to this as early as the recitals and that the period within which the Council has to make a decision - which is not clearly worded - is set at a maximum of three months .	Wir möchten sicherstellen , daß hierauf bereits in den Erwägungsgründen hingewiesen wird und die uneindeutig formulierte Frist , innerhalb der der Rat eine Entscheidung treffen muß , auf maximal drei Monate fixiert wird .	Quisiéramos asegurar que se aluda ya a esto en los considerandos y que el plazo , imprecisamente formulado , dentro del cual el Consejo ha de adoptar una decisión , se fije en tres meses como máximo .
2. Chapter 3, Färm (SV)	de	es
context Our experience of modern administration tells us that openness , decentralisation of responsibility and qualified evaluation are often as effective as detailed bureaucratic supervision .	Unsere Erfahrungen mit moderner Verwaltung besagen , daß Transparenz , Dezentralisation der Verantwortlichkeiten und eine qualifizierte Auswertung oft ebenso effektiv sind wie bürokratische Detailkontrolle .	Nuestras experiencias en materia de administración moderna nos señalan que la apertura , la descentralización de las responsabilidades y las evaluaciones bien hechas son a menudo tan eficaces como los controles burocráticos detallados .

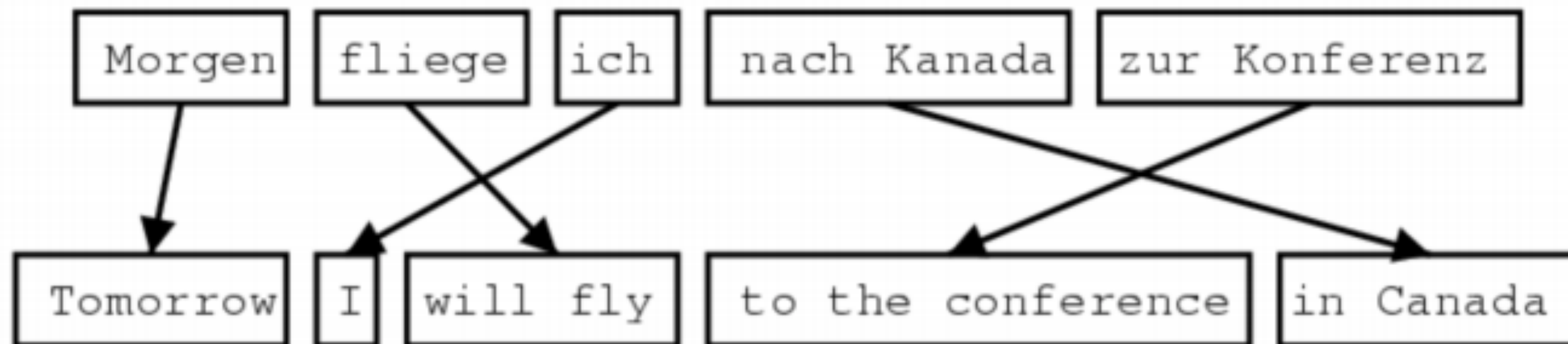
(Europarl, Koehn, 2005)

- Not available for many low-resource languages in the world

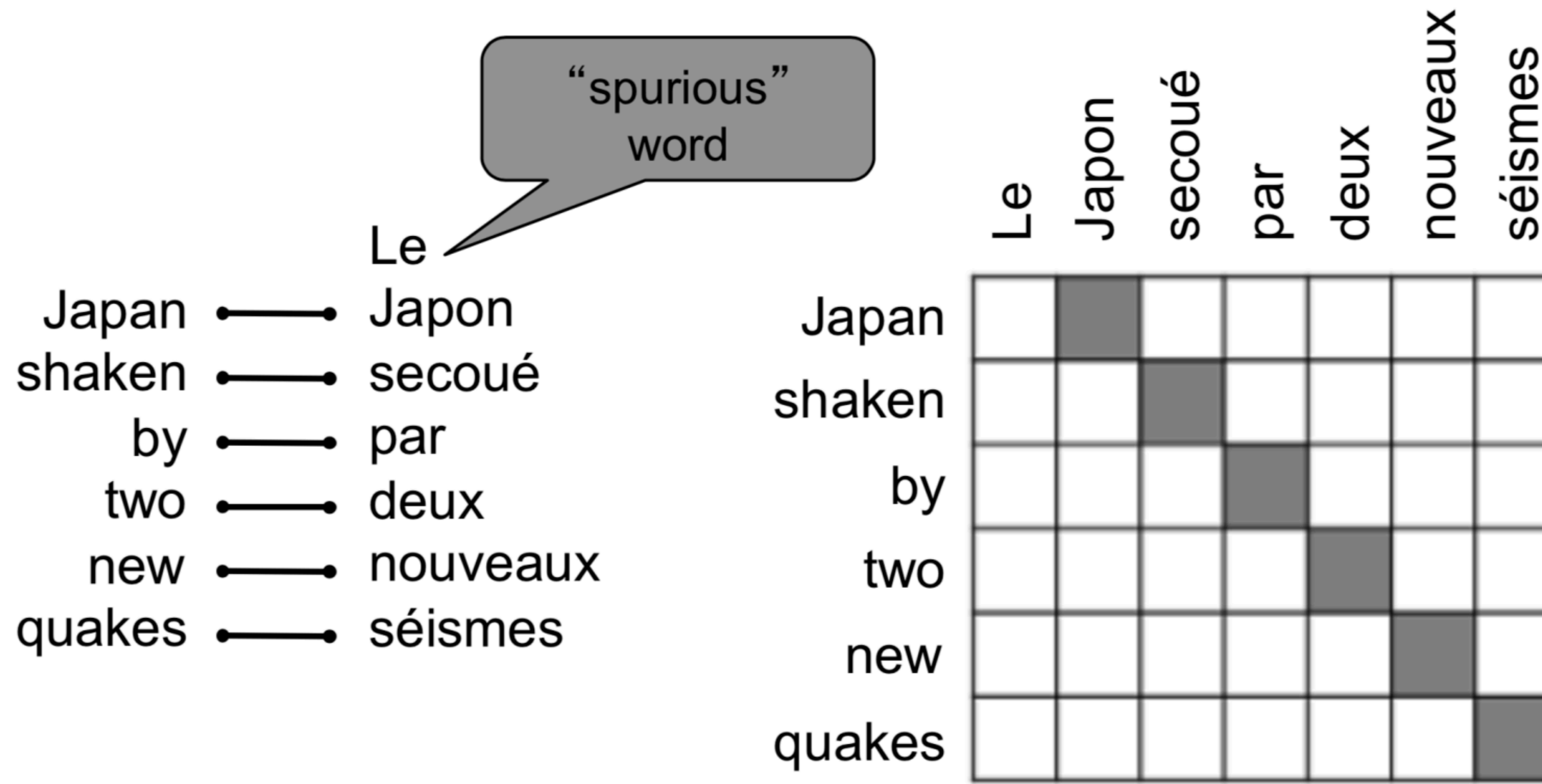
How to define the translation model?

Introduce latent variable modeling the **alignment** (word-level correspondence) between the source sentence **x** and the target sentence **y**

$$P(x | y) = P(x, A | y)$$



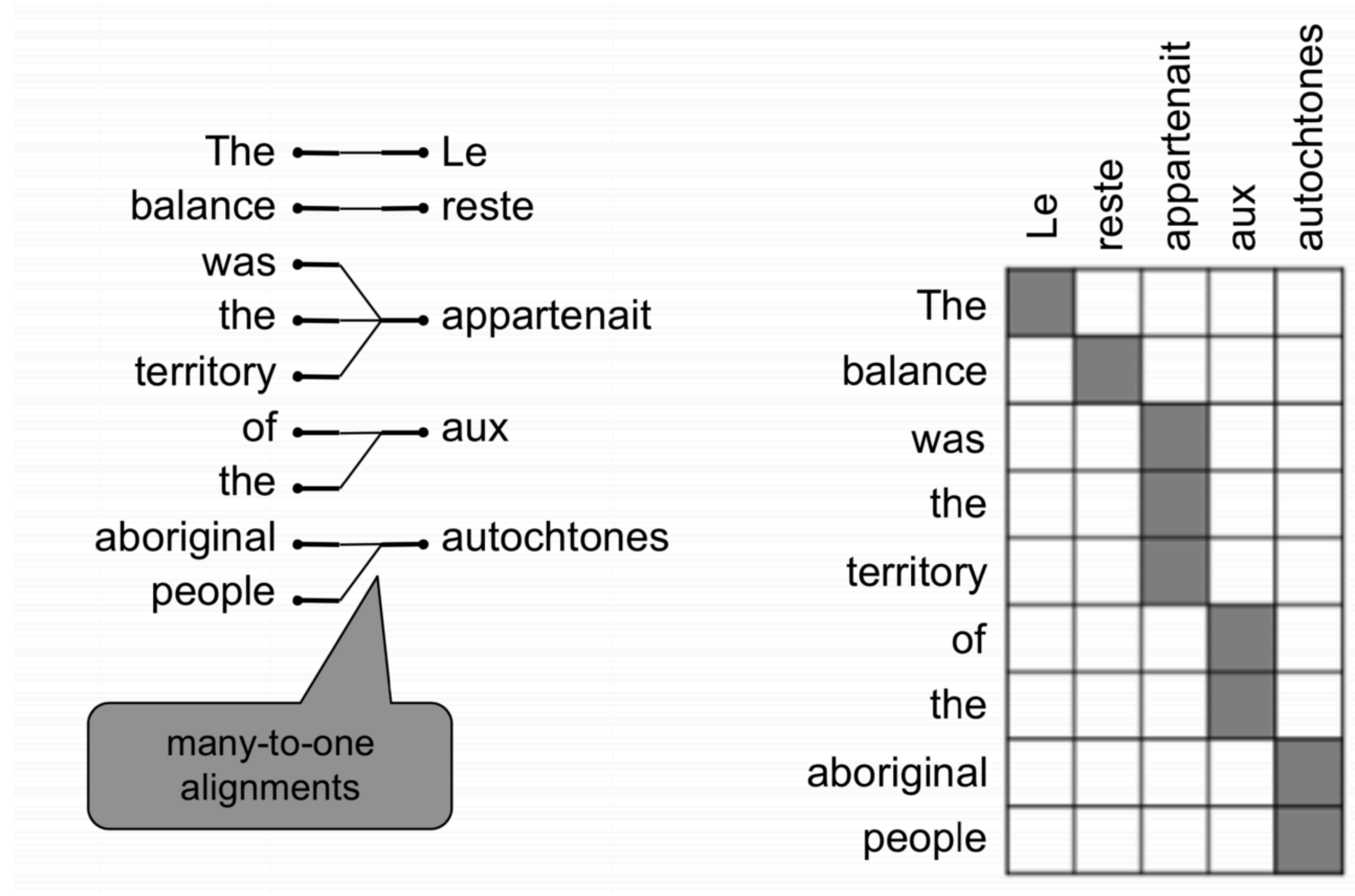
What is alignment?



Examples from: "The Mathematics of Statistical Machine Translation: Parameter Estimation", Brown et al, 1993. <http://www.aclweb.org/anthology/J93-2003>

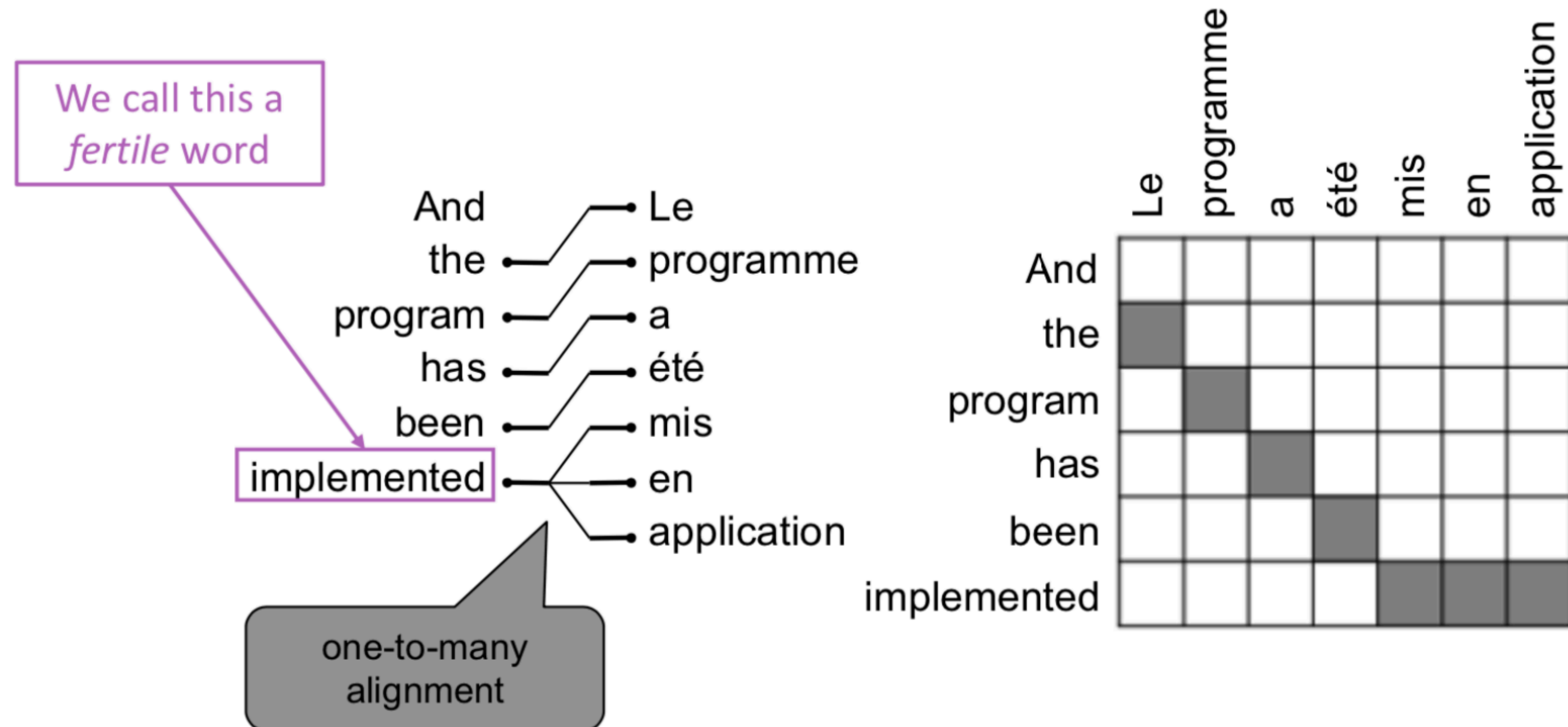
Alignment is complex

Alignment can be **many-to-one**



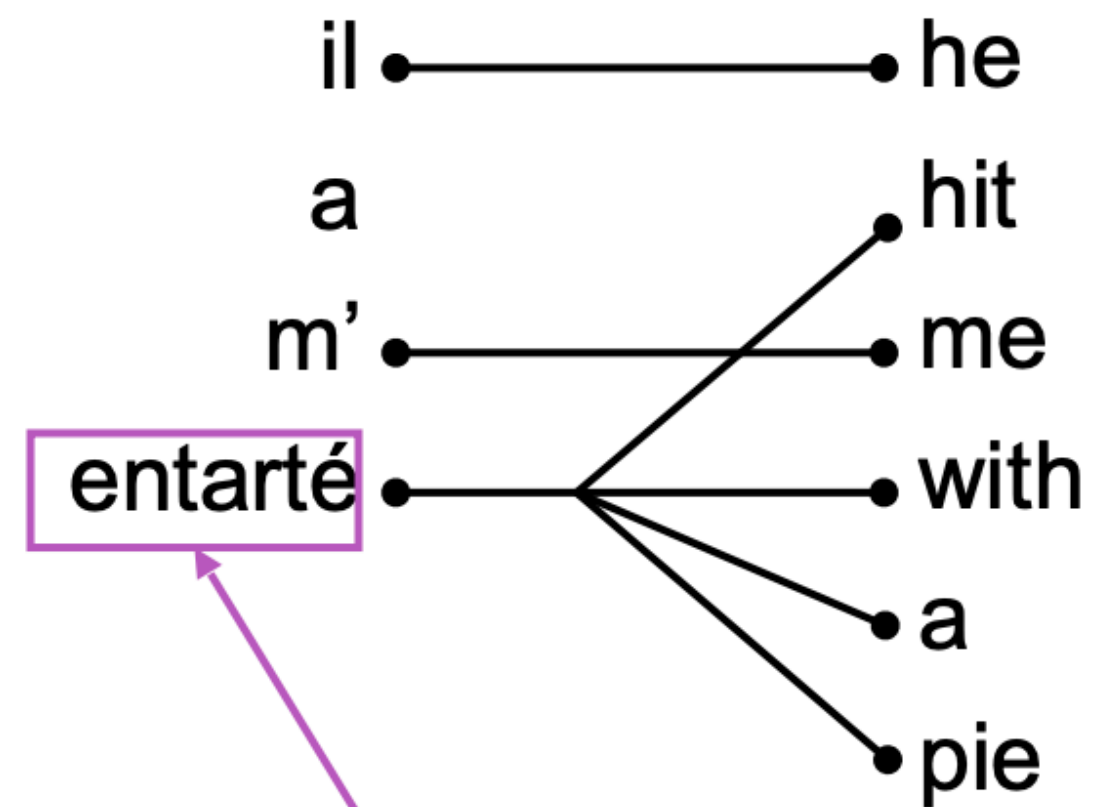
Alignment is complex

Alignment can be **one-to-many**



Alignment is complex

Some words are very fertile!

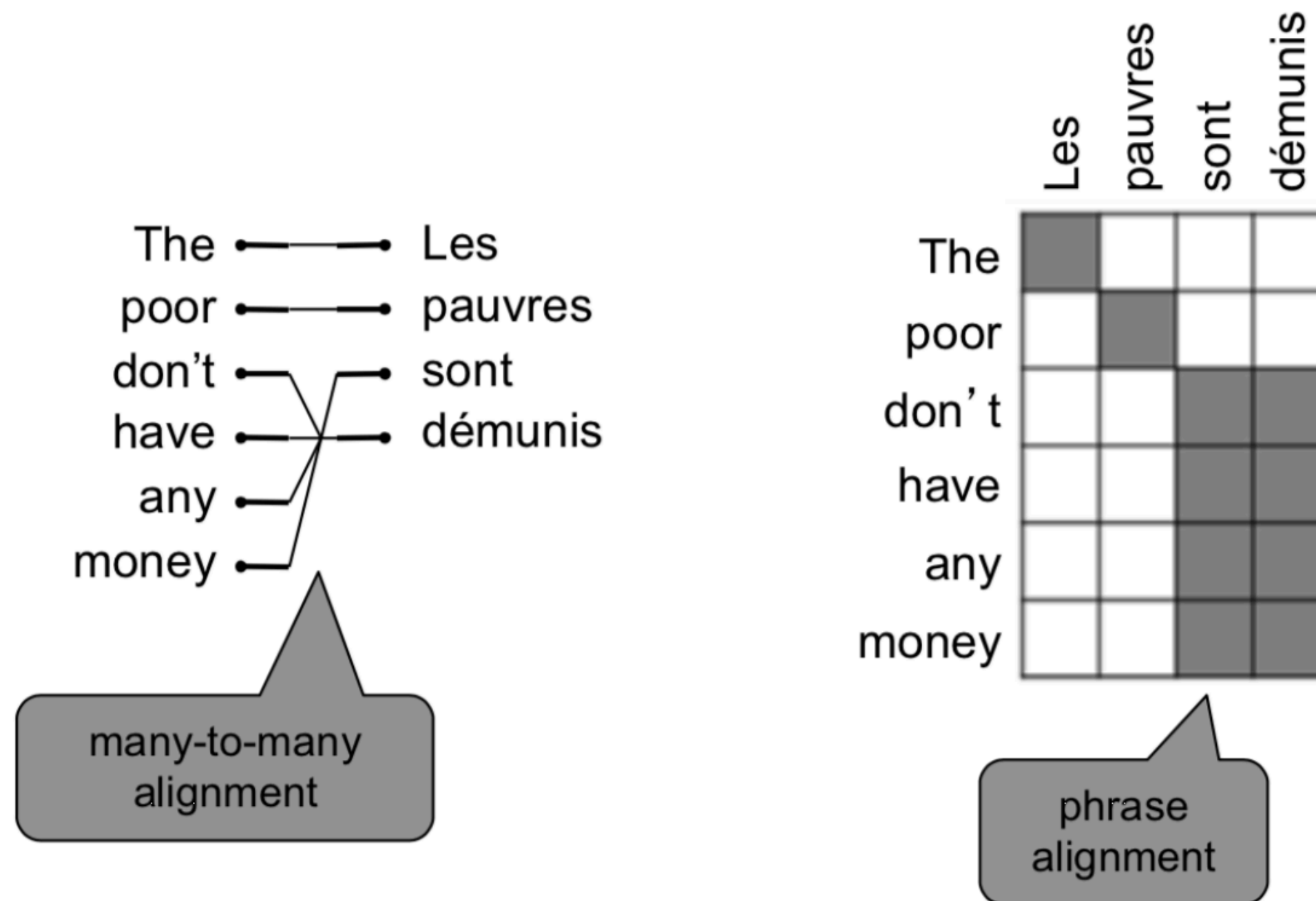


This word has no single-word equivalent in English

	he	hit	me	with	a	pie
il						
a						
m'						
entarté						

Alignment is complex

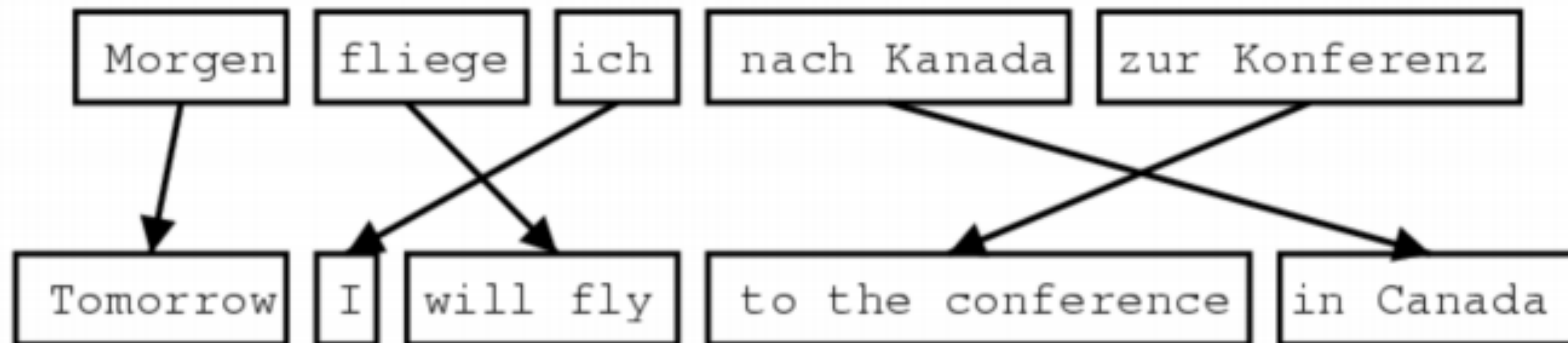
Alignment can be **many-to-many** (phrase-level)



How to define the translation model?

Given the alignment, how do we incorporate in our model?

$$P(x|y) = P(x, A|y)$$



Incorporating alignments

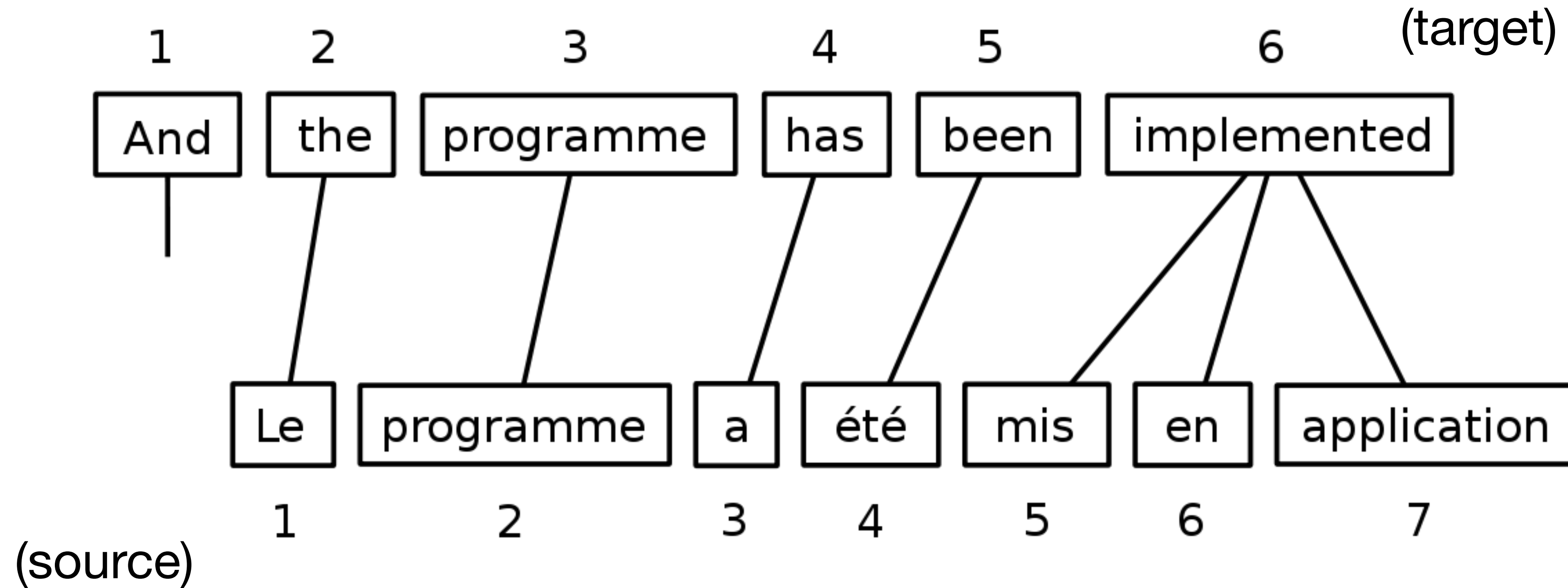
- Joint probability of alignment and translation can be defined as:

$$\begin{aligned} p(\mathbf{w}^{(s)}, \mathcal{A} \mid \mathbf{w}^{(t)}) &= \prod_{m=1}^{M^{(s)}} p(w_m^{(s)}, a_m \mid w_{a_m}^{(t)}, m, M^{(s)}, M^{(t)}) \\ &= \prod_{m=1}^{M^{(s)}} p(a_m \mid m, M^{(s)}, M^{(t)}) \times p(w_m^{(s)} \mid w_{a_m}^{(t)}). \end{aligned}$$

- $M^{(s)}, M^{(t)}$ are the number of words in source and target sentences
- a_m is the alignment of the m^{th} word in the source sentence, i.e. it specifies that the m^{th} word is aligned to the a_m^{th} word in target

Is this sufficient?

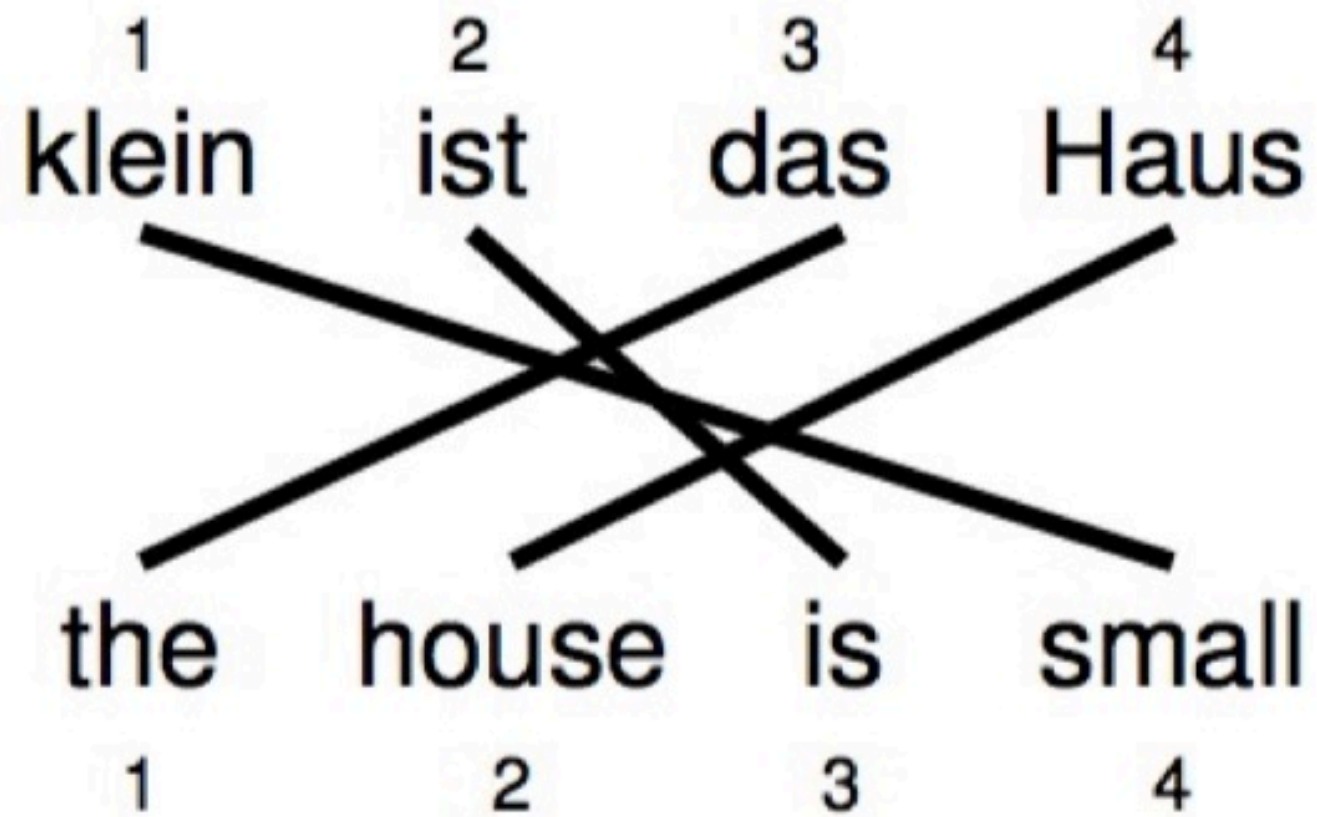
Incorporating alignments



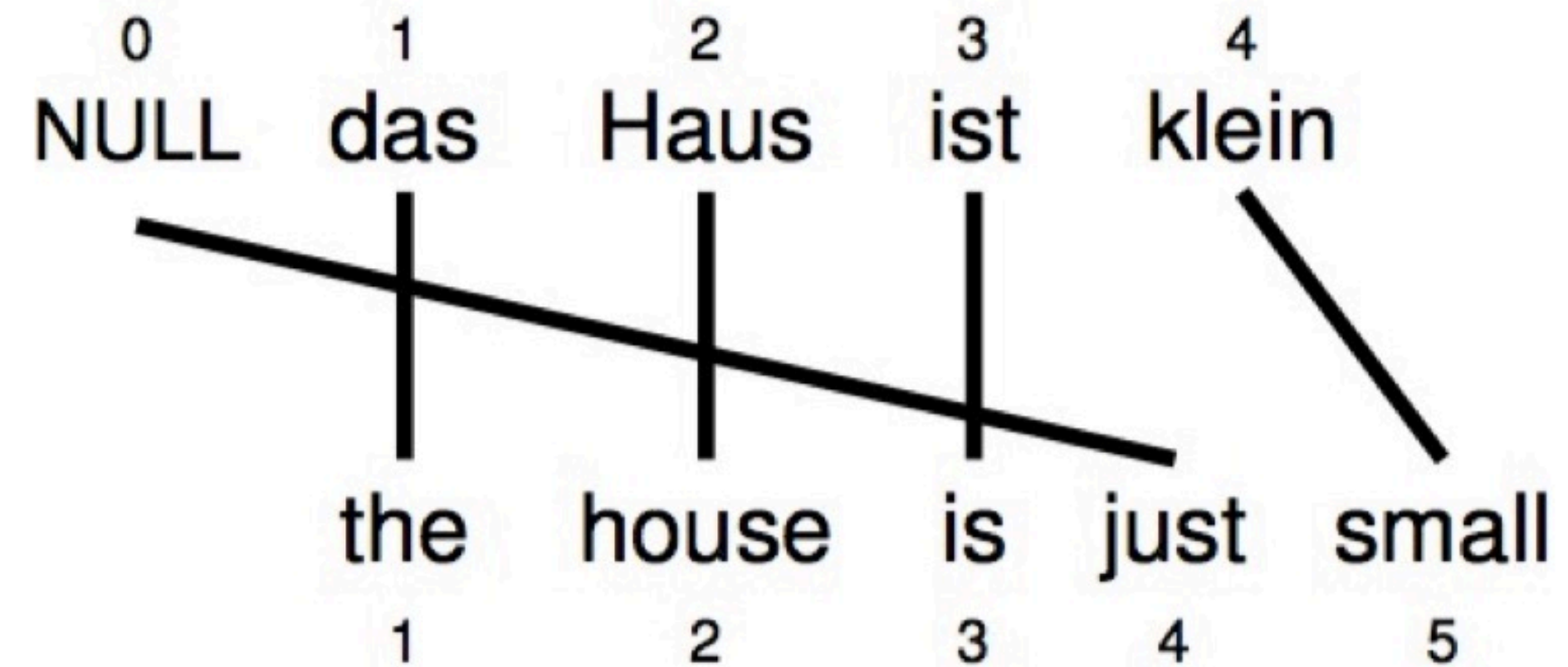
$$a_1 = 2, a_2 = 3, a_3 = 4, \dots$$

Multiple source words may align to the same target word!

Reordering and word insertion



$$\mathbf{a} = (3, 4, 2, 1)^\top$$



$$\mathbf{a} = (1, 2, 3, 0, 4)^\top$$

Assume extra NULL token

Independence assumptions

$$\begin{aligned} p(\mathbf{w}^{(s)}, \mathcal{A} \mid \mathbf{w}^{(t)}) &= \prod_{m=1}^{M^{(s)}} p(w_m^{(s)}, a_m \mid w_{a_m}^{(t)}, m, M^{(s)}, M^{(t)}) \\ &= \prod_{m=1}^{M^{(s)}} p(a_m \mid m, M^{(s)}, M^{(t)}) \times p(w_m^{(s)} \mid w_{a_m}^{(t)}). \end{aligned}$$

- Two independence assumptions:
 - Alignment probability factors across tokens:

$$p(\mathcal{A} \mid \mathbf{w}^{(s)}, \mathbf{w}^{(t)}) = \prod_{m=1}^{M^{(s)}} p(a_m \mid m, M^{(s)}, M^{(t)}).$$

- Translation probability factors across tokens:

$$p(\mathbf{w}^{(s)} \mid \mathbf{w}^{(t)}, \mathcal{A}) = \prod_{m=1}^{M^{(s)}} p(w_m^{(s)} \mid w_{a_m}^{(t)}),$$

How do we translate?

- We want: $\arg \max_{w^{(t)}} p(w^{(t)} | w^{(s)}) = \arg \max_{w^{(t)}} \frac{p(w^{(s)}, w^{(t)})}{p(w^{(s)})}$
- Sum over all possible alignments:

$$\begin{aligned} p(w^{(s)}, w^{(t)}) &= \sum_{\mathcal{A}} p(w^{(s)}, w^{(t)}, \mathcal{A}) \\ &= p(w^{(t)}) \sum_{\mathcal{A}} p(\mathcal{A}) \times p(w^{(s)} | w^{(t)}, \mathcal{A}) \end{aligned}$$

- Alternatively, take the max over alignments

Alignments

- **Key question:** How should we align words in source to words in target?

	<i>A</i>	<i>Vinay</i>	<i>le</i>	<i>gusta</i>	<i>python</i>
<i>Vinay</i>					
<i>likes</i>					
<i>python</i>					

good

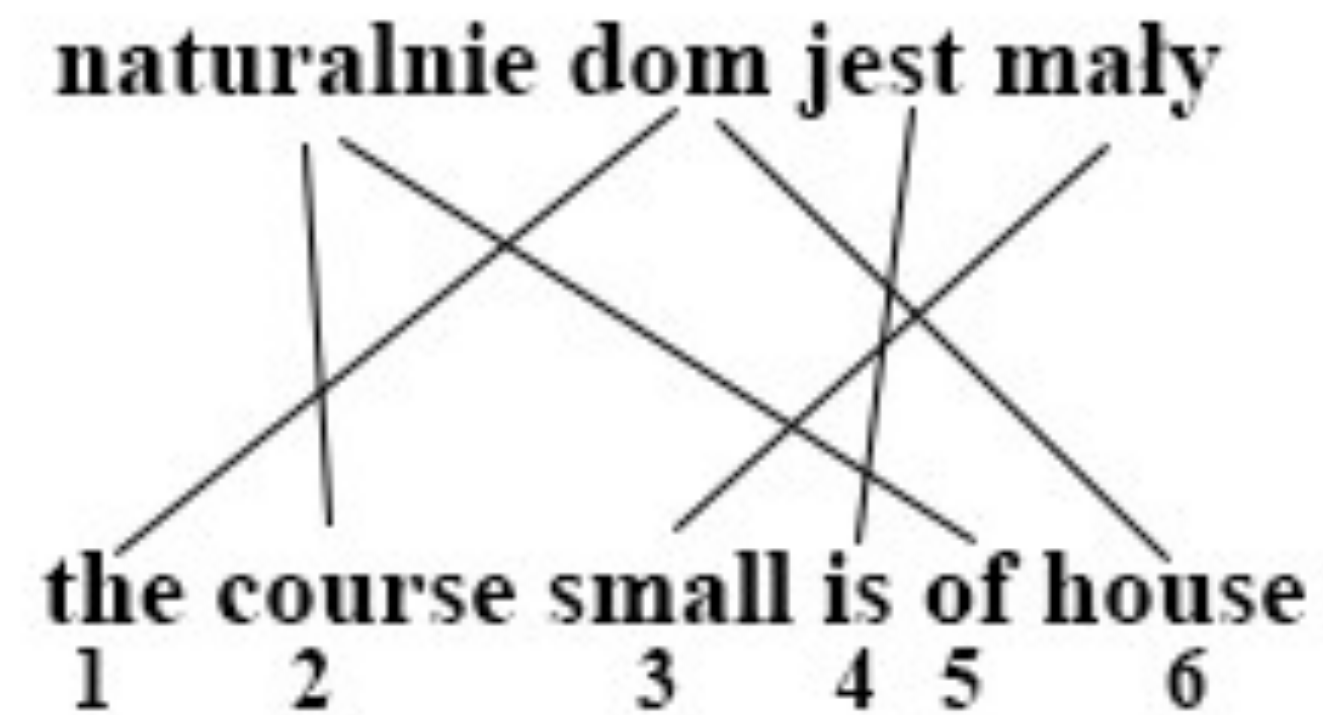
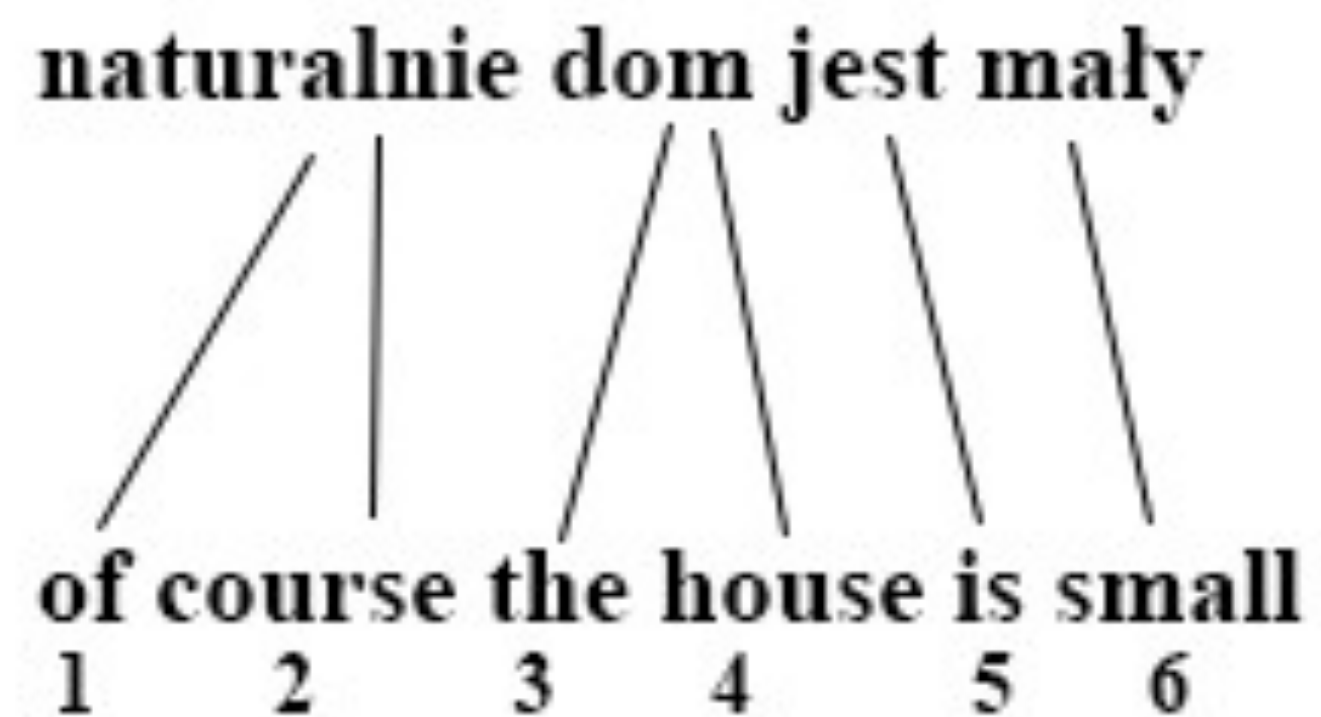
$$\mathcal{A}(\mathbf{w}^{(s)}, \mathbf{w}^{(t)}) = \{(A, \emptyset), (Vinay, Vinay), (le, likes), (gusta, likes), (Python, Python)\}.$$

bad

$$\mathcal{A}(\mathbf{w}^{(s)}, \mathbf{w}^{(t)}) = \{(A, Vinay), (Vinay, likes), (le, Python), (gusta, \emptyset), (Python, \emptyset)\}.$$

IBM Model I

- Assume $p(a_m | m, M^{(s)}, M^{(t)}) = \frac{1}{M^{(t)}}$
- Is this a good assumption?



Every alignment is equally likely!

IBM Model I

- Each source word is aligned to at most one target word

- Further, assume $p(a_m | m, M^{(s)}, M^{(t)}) = \frac{1}{M^{(t)}}$

- We then have:

$$p(w^{(s)}, w^{(t)}) = p(w^{(t)}) \sum_A \left(\frac{1}{M^{(t)}}\right)^{M^{(s)}} p(w^{(s)} | w^{(t)})$$

- How do we estimate $p(w^{(s)} = v | w^{(t)} = u)$?

IBM Model I

- If we had word-to-word alignments, we could compute the probabilities using the MLE:
 - $p(v | u) = \frac{\text{count}(u, v)}{\text{count}(u)}$
 - where $\text{count}(u, v) = \# \text{instances where word } u \text{ was aligned to word } v \text{ in the training set}$
- However, word-to-word alignments are often hard to come by

What can we do?

EM for Model I

- (E-Step) If we had an accurate translation model, we can estimate likelihood of each alignment as:

$$q_m(a_m \mid \mathbf{w}^{(s)}, \mathbf{w}^{(t)}) \propto p(a_m \mid m, M^{(s)}, M^{(t)}) \times p(w_m^{(s)} \mid w_{a_m}^{(t)}),$$

- (M Step) Use expected count to re-estimate translation parameters:

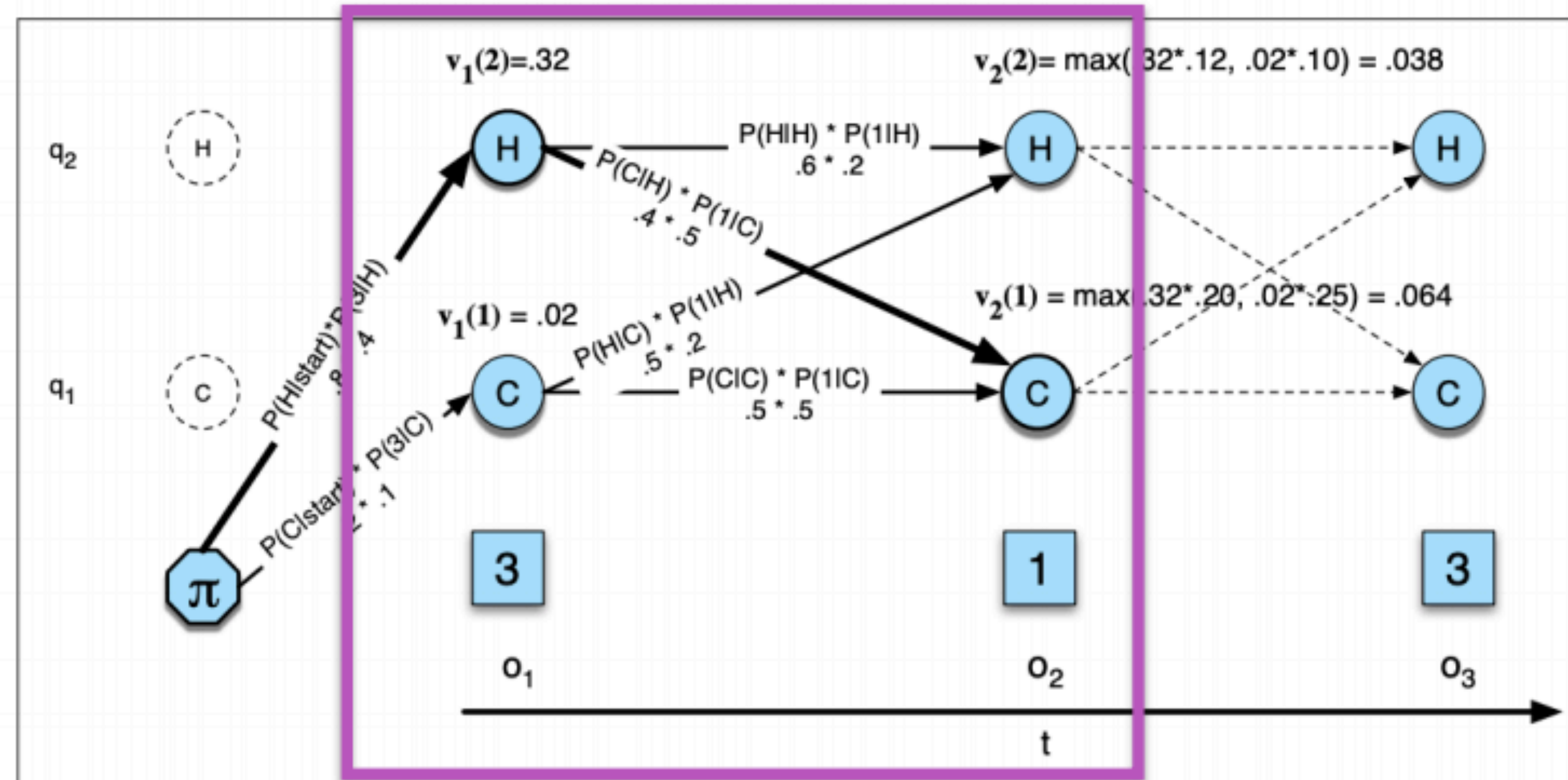
$$p(v \mid u) = \frac{E_q[\text{count}(u, v)]}{\text{count}(u)}$$

$$E_q [\text{count}(u, v)] = \sum_m q_m(a_m \mid \mathbf{w}^{(s)}, \mathbf{w}^{(t)}) \times \delta(w_m^{(s)} = v) \times \delta(w_{a_m}^{(t)} = u).$$

Independence assumptions allow for Viterbi decoding

Impose strong **independence assumptions** in model:

$$p(x, a|y) = \prod_{j=1}^{l_x} p(x_j | f_{a(j)})$$



Source: "Speech and Language Processing", Chapter A, Jurafsky and Martin, 2019.

In general, use greedy or beam decoding

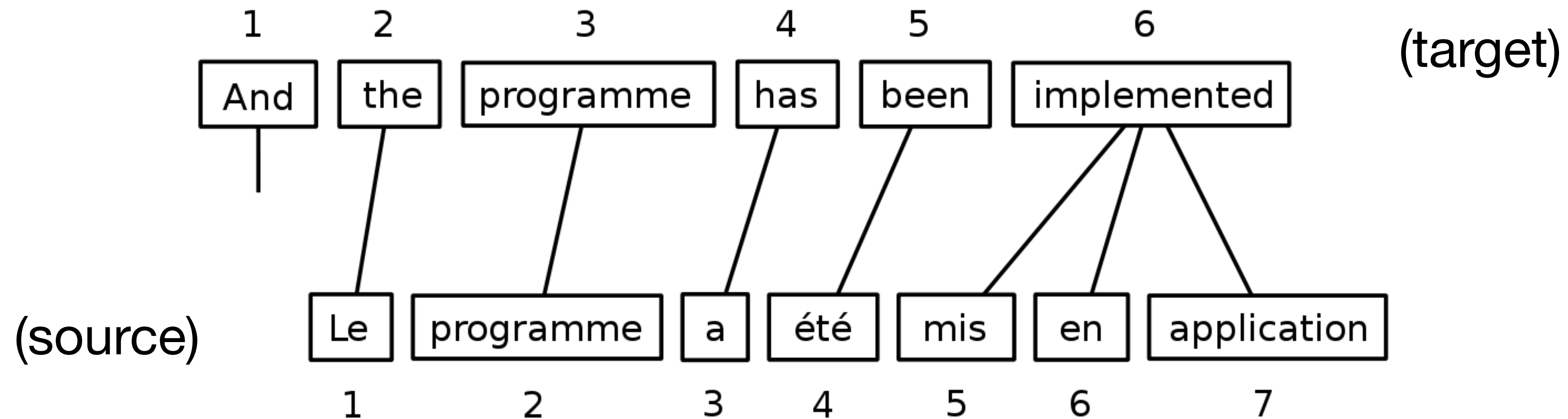
Model I: Decoding

- Pick target sentence length $M^{(t)}$
- Decode: $\arg \max_{w^{(t)}} p(w^{(t)} | w^{(s)}) = \arg \max_{w^{(t)}} p(w^{(s)}, w^{(t)})$

- $p(w^{(s)}, w^{(t)}) = p(w^{(t)}) \sum_A \frac{1}{M^{(t)}} p(w^{(s)} | w^{(t)})$

$$\begin{aligned} p(\mathbf{w}^{(s)}, \mathcal{A} | \mathbf{w}^{(t)}) &= \prod_{m=1}^{M^{(s)}} p(w_m^{(s)}, a_m | w_{a_m}^{(t)}, m, M^{(s)}, M^{(t)}) \\ &= \prod_{m=1}^{M^{(s)}} p(a_m | m, M^{(s)}, M^{(t)}) \times p(w_m^{(s)} | w_{a_m}^{(t)}). \end{aligned}$$

Model I: Decoding



At every step m , pick target word to maximize product of:

1. Language model: $p_{LM}(w_m^{(t)} | w_{<m}^{(t)})$

2. Translation model: $p(w_{b_m}^{(s)} | w_m^{(t)})$

where b_m is the inverse alignment from target to source

IBM Model 2

- Slightly relaxed assumption:
 - $p(a_m | m, M^{(s)}, M^{(t)})$ is also estimated, not set to constant
- Original independence assumptions still required:
 - Alignment probability factors across tokens:

$$p(\mathcal{A} | \mathbf{w}^{(s)}, \mathbf{w}^{(t)}) = \prod_{m=1}^{M^{(s)}} p(a_m | m, M^{(s)}, M^{(t)}).$$

- Translation probability factors across tokens:

$$p(\mathbf{w}^{(s)} | \mathbf{w}^{(t)}, \mathcal{A}) = \prod_{m=1}^{M^{(s)}} p(w_m^{(s)} | w_{a_m}^{(t)}),$$

Other IBM models

Model 1: lexical translation

Model 2: additional absolute alignment model

Model 3: extra fertility model

Model 4: added relative alignment model

Model 5: fixed deficiency problem.

Model 6: Model 4 combined with a [HMM](#) alignment model in a log linear way

- Models 3 - 6 make successively weaker assumptions
 - But get progressively harder to optimize
- Simpler models are often used to ‘initialize’ complex ones
 - e.g train Model 1 and use it to initialize Model 2 parameters

Phrase-based MT

- Word-by-word translation is not sufficient in many cases

Nous allons prendre un verre

(literal)

We will take a glass

(actual)

We'll have a drink

- Solution: build alignments and translation tables between multiword spans or “phrases”

	<i>Nous</i>	<i>allons</i>	<i>prendre</i>	<i>une</i>	<i>verre</i>
We'll					
have					
a					
drink					

Phrase-based MT

- Solution: build alignments and translation tables between multiword spans or “phrases”
- Translations condition on multi-word units and assign probabilities to multi-word units
- Alignments map from spans to spans

$$p(\mathbf{w}^{(s)} \mid \mathbf{w}^{(t)}, \mathcal{A}) = \prod_{((i,j),(k,\ell)) \in \mathcal{A}} P_{\mathbf{w}^{(s)} \mid \mathbf{w}^{(t)}}(\{w_{i+1}^{(s)}, w_{i+2}^{(s)}, \dots, w_j^{(s)}\} \mid \{w_{k+1}^{(t)}, w_{k+2}^{(t)}, \dots, w_\ell^{(t)}\})$$

Syntactic MT

- ▶ Rather than use phrases, use a *synchronous context-free grammar*: constructs “parallel” trees in two languages simultaneously

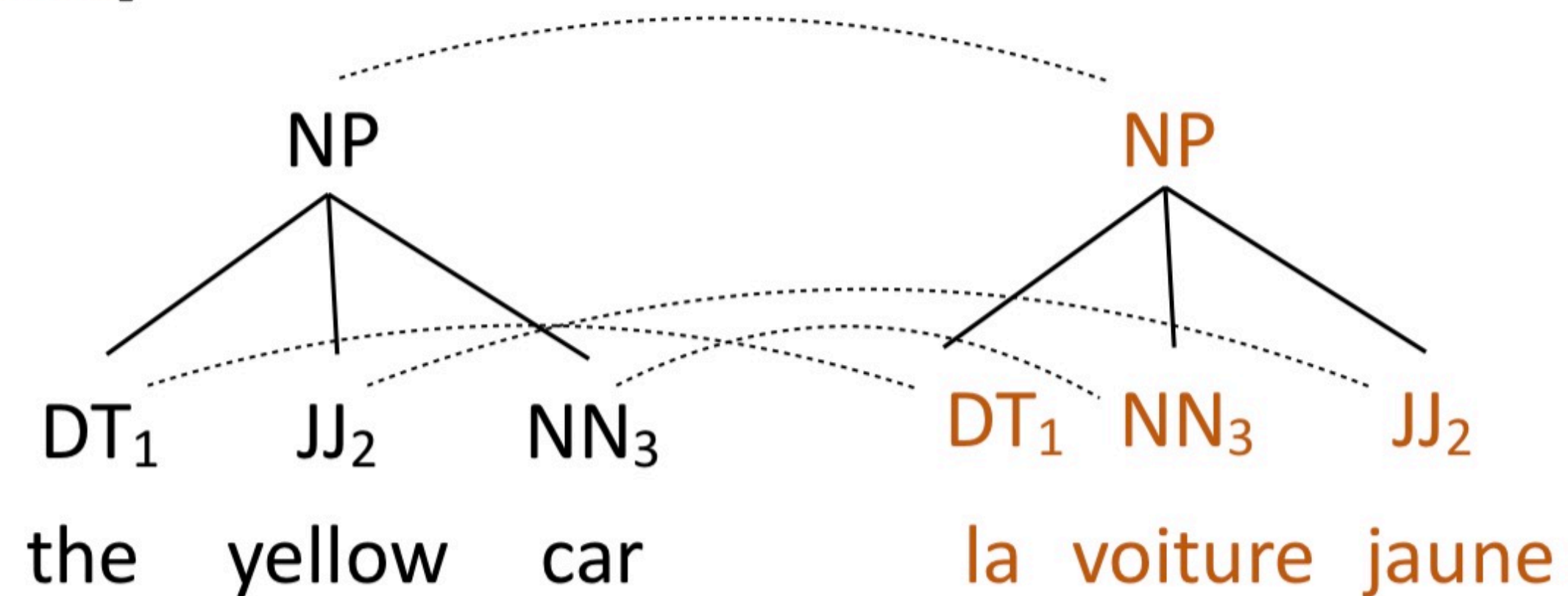
NP $\rightarrow$ [DT₁ JJ₂ NN₃; DT₁ NN₃ JJ₂]

DT $\rightarrow$ [the, la]

DT $\rightarrow$ [the, le]

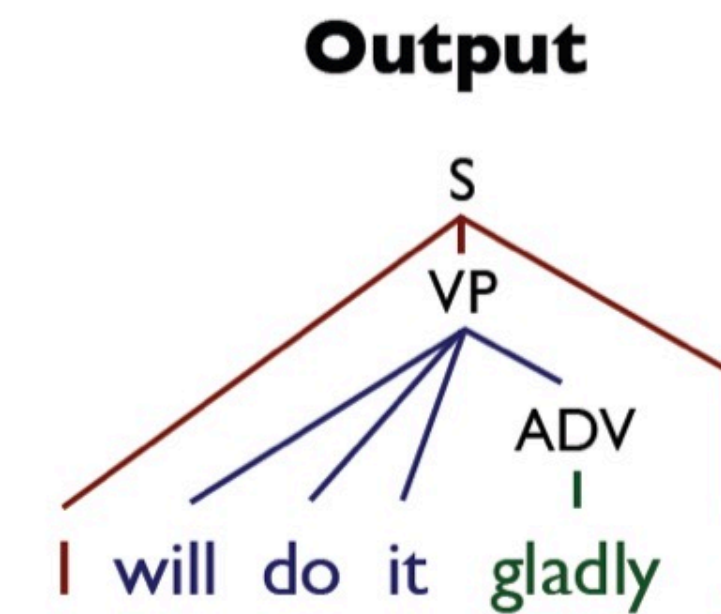
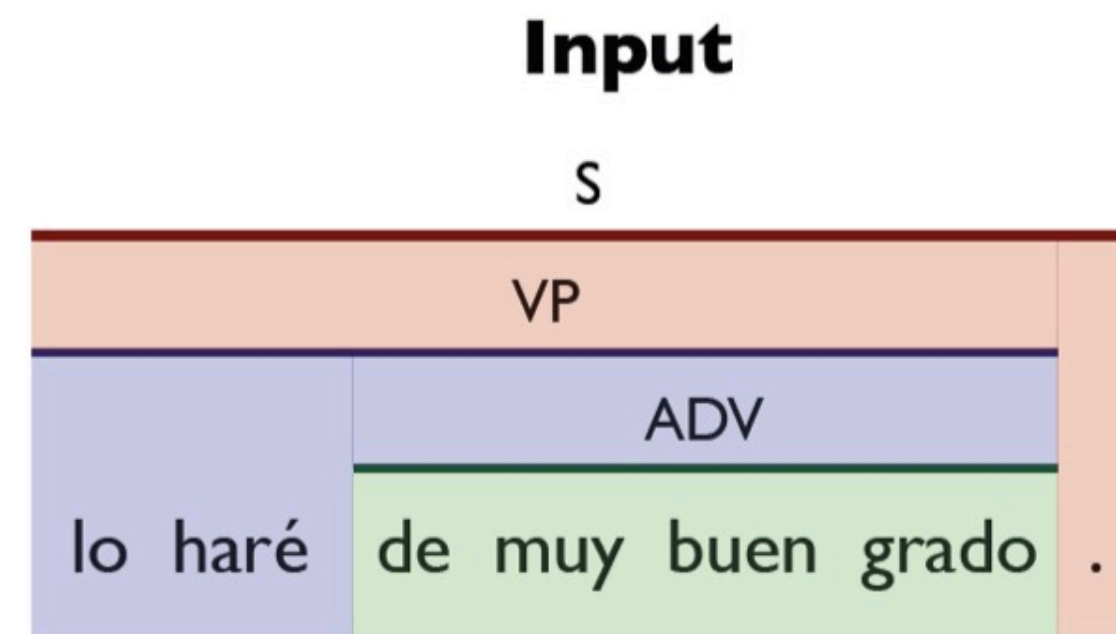
NN $\rightarrow$ [car, voiture]

JJ $\rightarrow$ [yellow, jaune]



- ▶ Assumes parallel syntax up to reordering
- ▶ Translation = parse the input with “half” the grammar, read off other half

Syntactic MT



Grammar

- ▶ Relax this by using lexicalized rules, like “syntactic phrases”
- ▶ Leads to HUGE grammars, parsing is slow

$S \rightarrow \langle VP . ; I VP . \rangle$ **OR** $S \rightarrow \langle VP . ; you VP . \rangle$

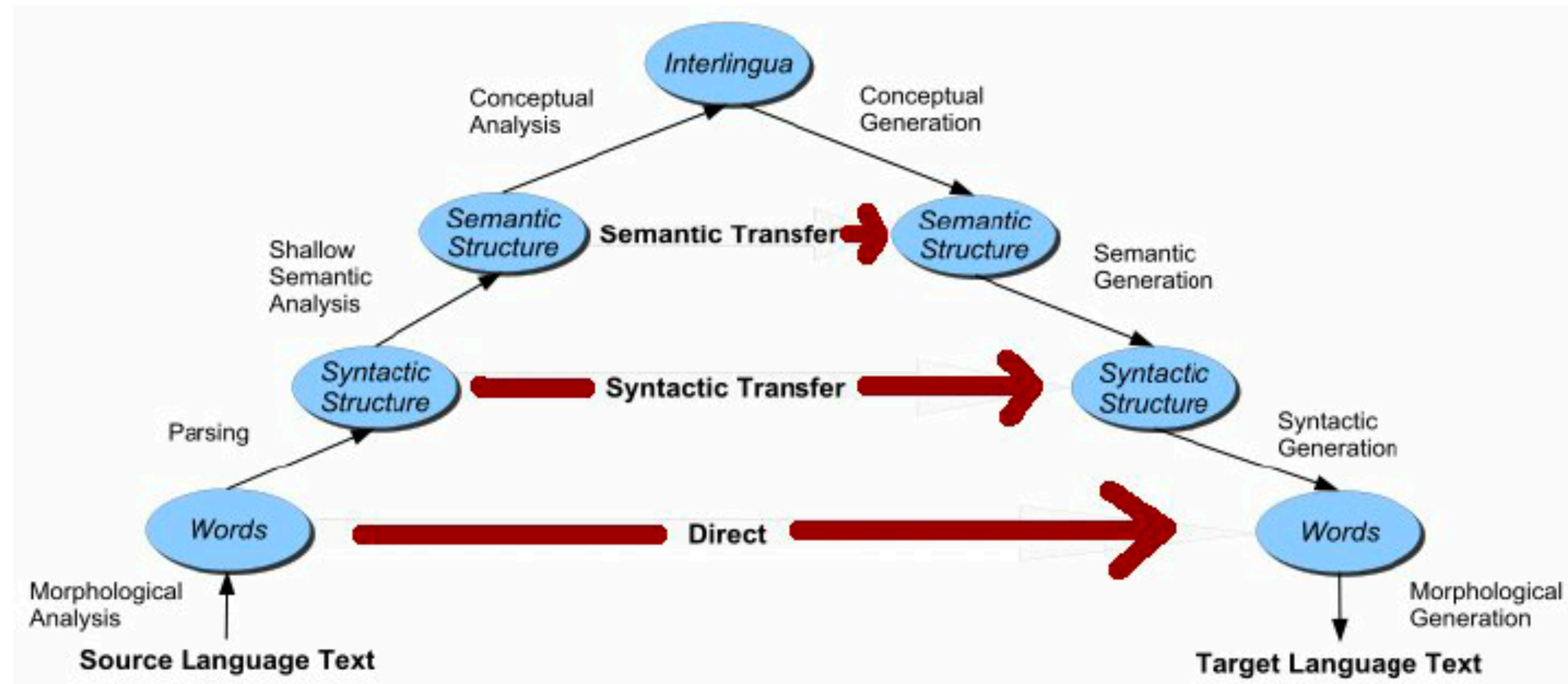
$VP \rightarrow \langle lo haré ADV ; will do it ADV \rangle$

$s \rightarrow \langle lo haré ADV . ; I will do it ADV . \rangle$

$ADV \rightarrow \langle de muy buen grado ; gladly \rangle$

Slide credit: Dan Klein

Vauquois Pyramid



- Hierarchy of concepts and distances between them in different languages
- Lowest level: individual words/characters
- Higher levels: syntax, semantics
- Interlingua: Generic language-agnostic representation of meaning

Statistical MT

- 1990s to 2010s: huge research area
- Extremely complex systems
 - Many separately-designed subcomponents
 - Lots of feature engineering
 - Needed to compile/maintain extra resources (phrase tables)
- Lots of human effort to maintain

History

- Started in the 1950s: rule-based, tightly linked to formal linguistics theories
- 1980s: Statistical MT
- 2000s-2015: Statistical Phrase-Based MT
- 2015-Present: **Neural Machine Translation**
- ~2018-Present: **Neural Machine Translation + PBMT Hybrid**